Université Paris-Saclay / ENSIIE

Master 2 — Risk and Asset Management

Project – Volatility Models

Calibration of SABR and Heston Models on SPX and VIX Options

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1. Introduction

Modeling implied volatility remains a central issue in quantitative finance, both for pricing and for risk management. In practice, option markets display persistent deviations from the Black–Scholes framework, notably through volatility smiles, term structures, and cross-market consistency issues between equity and volatility derivatives. A relevant volatility model should therefore not only fit equity index options accurately, but also remain coherent when confronted with products linked to volatility itself.

The purpose of this project is to compare two classical stochastic volatility frameworks, namely the SABR model (Hagan et al., 2002) and the Heston model (Heston, 1993), on U.S. equity and volatility option data. More precisely, we study their ability to reproduce the implied volatility structure of SPX call options and to explain VIX call prices and implied volatilities within a unified calibration framework. The analysis is conducted on a fixed market snapshot and relies exclusively on out-of-the-money call options, in accordance with the implementation retained in the final notebook.

The project is organized around three main objectives. First, we calibrate both models to SPX options and assess their ability to reproduce the observed smile structure. Second, we extend the analysis to VIX options in order to evaluate whether a model calibrated on equity options can also account for volatility derivatives, either in a standard calibration setting or in a joint SPX–VIX calibration procedure. Third, we investigate the path regularity of the simulated processes through an empirical estimation of the Hölder exponent, in order to relate the numerical behavior of the models to their theoretical diffusion structure.

The comparison between SABR and Heston is particularly relevant in this project because the two models are implemented and calibrated in different ways, both on SPX and VIX options. In our framework, SABR is used with the Hagan implied volatility approximation for SPX options and with Monte Carlo pricing for VIX options, whereas Heston is calibrated with the analytic QuantLib engine for SPX options and combined with the exact analytical relation between the VIX and future variance for VIX pricing. This numerical setup makes it possible to compare not only the quality of fit on SPX options, but also the ability of each framework to incorporate VIX information through either a standard calibration or a joint SPX–VIX calibration.

The final results show a clear trade-off between local fit quality and cross-market coherence. SABR in its standard version provides the best SPX fit, whereas its joint version substantially improves the VIX fit at the expense of a noticeable deterioration on SPX. Heston, although less accurate than SABR on SPX alone, delivers a more balanced compromise once VIX information is introduced. The empirical regularity analysis also confirms that all retained processes remain consistent with standard Brownian diffusion behavior, with estimated Hölder exponents close to 0.5.

The remainder of the report is organized as follows. Section 2 presents the market data, the preprocessing choices, and the construction of the fixed snapshot used throughout the study. Section 3 is devoted to the SABR model and its standard and joint calibrations. Section 4 presents the Heston model and its calibration methodology. Section 5 studies the empirical Hölder regularity of the simulated processes. Section 6 compares the models globally in terms of SPX fit, VIX fit, and economic interpretation, before a final discussion of the main limitations of the study.

2. Market Data and Preprocessing

The empirical analysis is based on U.S. equity and volatility option data obtained through yfinance, using a fixed market snapshot in order to ensure consistency across all computations. In the final implementation, the snapshot is first collected once and then stored in a single CSV file, which is subsequently reloaded throughout the notebook. This choice guarantees that all calibrations, comparisons, and regularity analyses are performed on exactly the same market inputs. In particular, the spot levels of the SPX and the VIX, as well as the short-term interest rate proxy, are frozen at the snapshot date and reused consistently in all subsequent sections.

Parameter	**Value**	**Description**
Snapshot date	2026-04-09	Fixed date used throughout
SPX spot	6,831.35	S&P 500 index level
VIX spot	19.75	Proxy for VIX futures underlying
Risk-free rate r	3.58%	Short-term rate proxy
Dividend yield q	1.50%	Index continuous dividend yield
SPX OTM calls	1,734	Contracts retained after filtering
VIX OTM calls	274	Contracts retained after filtering (6 maturities)

Table A. Market snapshot used throughout the study (source: yfinance, date 2026-04-09).

The study focuses exclusively on out-of-the-money call options. For SPX options, this means retaining call contracts with strikes above the current spot level of the index. The same logic is applied to VIX options. This restriction is consistent with the final notebook and was adopted in order to work with a homogeneous dataset and avoid mixing option classes with potentially different liquidity profiles and smile behaviors. All time-to-maturity variables are expressed in years, using calendar days divided by 365, which is also consistent with the implementation used for pricing and calibration.

Several preprocessing steps are applied before calibration. First, raw option chains are downloaded across available maturities within a predefined maturity window. For each retained contract, the mid-price is computed as the average of bid and ask quotes. Implied volatilities are then recovered by inverting the Black–Scholes call pricing formula. Observations for which implied volatility inversion fails, or produces unrealistic values, are discarded. Additional liquidity filters are also imposed. For SPX, the filtering is relatively strict, reflecting the fact that SPX options are highly liquid and that it is therefore reasonable to remove contracts with poor quotes. For VIX options, the filtering is deliberately more flexible, since quoted spreads are often wider and the market is less homogeneous. This asymmetry in the preprocessing stage follows directly from the final notebook and is justified by the different microstructure properties of the two markets.

The retained moneyness ranges also differ between SPX and VIX. For SPX, the sample is restricted to a relatively narrow region around the traded smile in order to keep the dataset focused on the part of the market that is most informative for model calibration. For VIX, the admissible moneyness interval is wider, reflecting the larger dispersion of quoted strikes in the volatility options market. After this first filtering stage, an additional SPX subsample is defined for the Heston calibration, with moneyness restricted to the interval used in the final code. This choice is motivated by numerical stability and by the objective of concentrating the Heston fit on the most relevant part of the SPX smile.

A final practical issue concerns the treatment of VIX options. In the notebook, Black–Scholes implied volatilities for VIX calls are computed by using the spot VIX as underlying input. Strictly speaking, listed VIX options are written on VIX futures rather than on the VIX spot index. Since yfinance does not provide a directly integrated and convenient VIX futures curve in the framework retained for this project, the spot VIX is used as a proxy. This simplification must be kept in mind when interpreting VIX implied volatilities and the associated RMSE values. It does not affect the internal consistency of the numerical comparison within our notebook, but it remains a limitation of the market data construction.

In order to visualize the structure of the retained market sample, we first plot the implied volatility smiles of OTM SPX call options across maturities. The figure shows a rich cross-section of smiles with a strong maturity dimension. The SPX dataset is large and covers many maturities, which makes it suitable for testing both slice-by-slice fitting accuracy and cross-maturity behavior. The figure also illustrates that, even after filtering, the market data remain heterogeneous across maturities, which justifies the use of flexible calibration procedures.

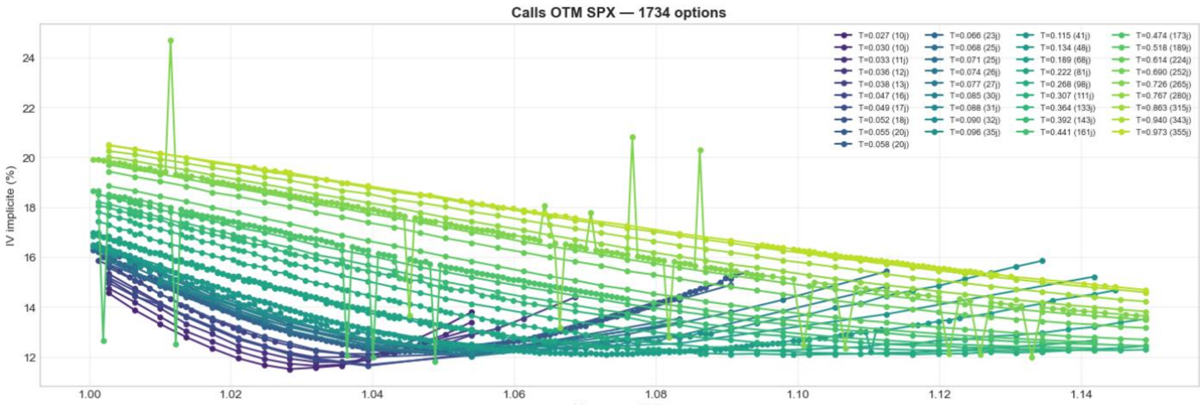


Figure 1. Market implied volatility smiles for out-of-the-money SPX call options across maturities after preprocessing.

We then consider the retained VIX call implied volatilities. Compared with SPX, the VIX dataset is smaller and noisier, but it still provides meaningful information on the volatility-derivative market. The smiles are observed over a smaller set of maturities and display much higher implied volatility levels, which is consistent with the nature of VIX options. This second figure is particularly important because the project does not stop at fitting SPX options alone: one of the main objectives is precisely to evaluate whether information from VIX options can be incorporated through a joint calibration procedure.

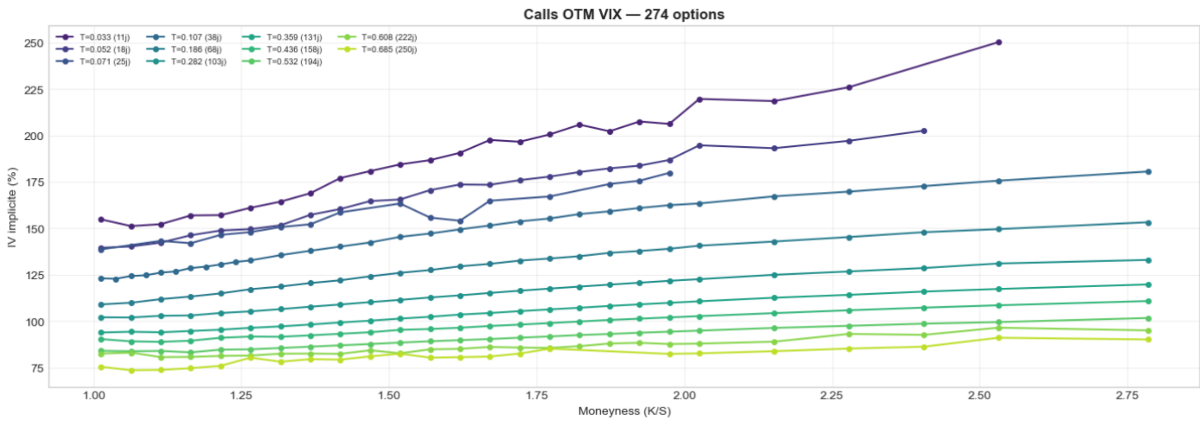


Figure 2. Market implied volatility smiles for out-of-the-money VIX call options across maturities after preprocessing.

Overall, the preprocessing stage produces two coherent datasets that serve distinct but complementary purposes. The SPX sample provides the main benchmark for fitting the equity implied volatility structure, while the VIX sample makes it possible to test how well each model can accommodate volatility-market information in addition to equity-option data. These two datasets form the empirical basis of the calibration and comparison exercises developed in the following sections.

3. SABR Model

In the first modeling stage of the project, we consider the SABR framework (Hagan et al., 2002) as a flexible parametric model for reproducing the implied volatility structure of SPX call options and, in a second step, for incorporating information from VIX call options. In its standard form, the SABR dynamics are written as

$$dF_t = \sigma_t F_t^{\beta} dW_t^{(1)}, \qquad d\sigma_t = \alpha \sigma_t dW_t^{(2)}, \qquad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho\, dt,$$

where $F_t$denotes the forward, $\sigma_t$the instantaneous volatility process, $\beta$the elasticity parameter, $\alpha$the volatility-of-volatility coefficient, and $\rho$the instantaneous correlation between the two Brownian motions. In practice, the model is used here as an implied-volatility fitting tool for SPX options through the Hagan approximation, and as a simulation-based model for VIX options.

For SPX options, the implementation relies on the robust Hagan implied volatility approximation (Hagan et al., 2002). For each maturity, the model parameters $(\sigma_0, \alpha, \rho)$are calibrated slice by slice, while $\beta$is treated separately and selected globally. The forward level used in each slice is computed from the spot index and the deterministic carry adjustment based on the short rate and dividend yield retained in the data section. The objective is to minimize the squared deviation between market implied volatilities and model-implied volatilities across strikes. In order to improve numerical stability, the calibration uses a pragmatic warm-start strategy: a global optimization is first performed on the first maturity slice, and the resulting parameters are then used as initialization for the following maturities through a local constrained optimization routine. This procedure is consistent with the final notebook and is designed to reduce computation time while preserving a stable maturity-by-maturity fit.

A key preliminary step concerns the choice of the elasticity parameter β. Rather than fixing it arbitrarily, the notebook evaluates a grid of candidate values between 0 and 1 and selects the value minimizing the global SPX fitting error. In the final retained implementation, the best value is obtained at $\beta^* = 0.7$, which yields the lowest overall RMSE on the SPX dataset. This step is important because βstrongly influences the shape of the smile and therefore the balance between low-strike and high-strike fit.

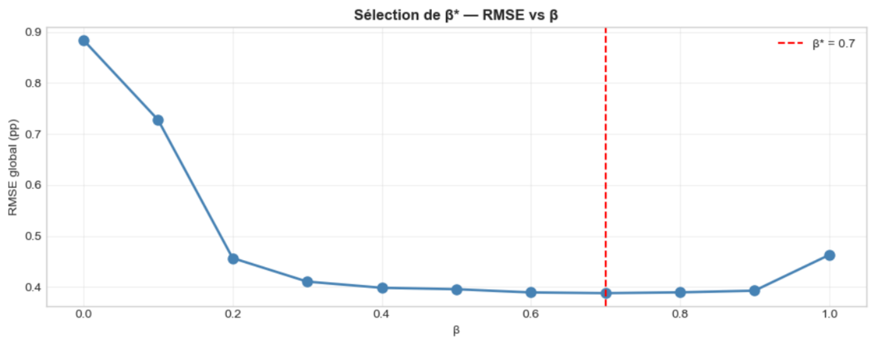


Figure 3. Global SPX RMSE as a function of the SABR elasticity parameter β. The minimum is obtained at $\beta^* = 0.7$, which is retained in the rest of the SABR analysis.

β	RMSE SPX (pp)
0.0	0.8844

β	RMSE SPX (pp)
0.1	0.7279
0.2	0.4567
0.3	0.4105
0.4	0.3984
0.5	0.3956
0.6	0.3893
0.7 *	0.3879
0.8	0.3894
0.9	0.3927
1.0	0.4628

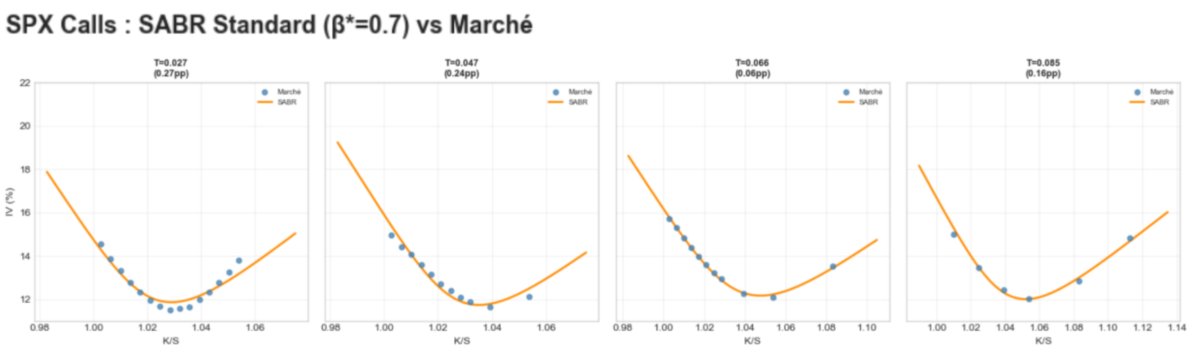
Table B. SABR SPX RMSE as a function of β across the full grid. The asterisk (*) marks the retained value $\beta^* = 0.7$.

Since SABR is calibrated maturity by maturity, it does not produce a single global parameter vector. Table 1 reports representative calibrated SABR parameters for the maturity slice later used in the regularity analysis.

Calibration	(T_{\mathrm{rep}})	(\beta)	(\sigma_0)	(\alpha)	(\rho)
SABR Standard	0.096	0.7	2.292896	3.250182	-0.732310
SABR Joint	0.096	0.7	2.187513	1.741347	-0.626193

Table 1. Representative calibrated SABR parameters

Once β is fixed, the standard SABR calibration produces a very accurate fit of SPX implied volatilities. The figure below shows that the model is able to reproduce the shape of the observed SPX smiles across a wide range of maturities. This result is one of the main strengths of the SABR specification in the present project: because the model is calibrated independently by maturity, it adapts very well to the cross-sectional geometry of the equity implied volatility surface. In the final notebook, this strong local fitting ability is reflected by the best SPX RMSE among all retained model configurations.



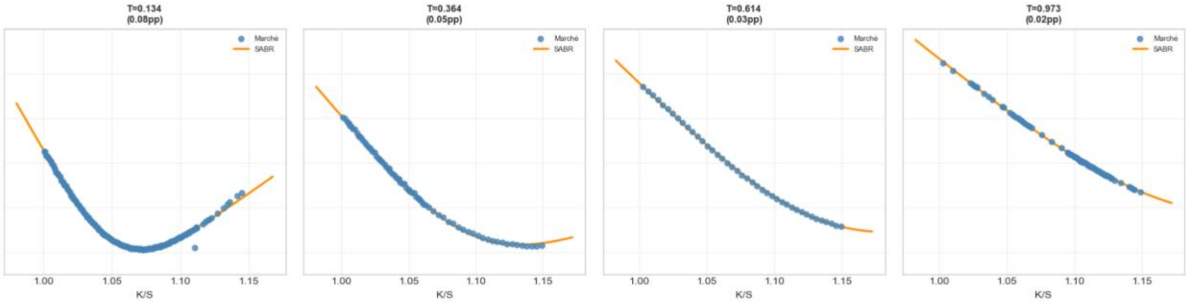


Figure 4. Market and model implied volatility smiles for SPX out-of-the-money call options under the standard SABR calibration for selected maturities.

The slice-by-slice nature of the calibration, however, also reveals one of the main limitations of the approach. Since each maturity is treated almost independently, the model does not impose a truly coherent intertemporal dynamic structure across the full surface. In other words, SABR performs very well as a cross-sectional fitting device, but it is less naturally suited to capturing a unified time-consistent stochastic volatility dynamics across maturities. This limitation becomes more apparent when one goes beyond SPX options and attempts to incorporate VIX option information in the calibration.

The next step consists in evaluating the SABR model on VIX call options. In the notebook, VIX options are not priced through a closed-form approximation. Instead, they are evaluated by Monte Carlo simulation of the SABR volatility dynamics over the relevant horizon. The VIX level at option maturity is approximated from the future integrated variance over a 30-day window, in a way that is consistent with the numerical structure retained in the implementation. This simulation-based approach makes it possible to extend the SABR framework beyond SPX smiles, but it also introduces additional numerical complexity and, more importantly, exposes a conceptual weakness of the model in the present setting: a specification that is highly efficient for fitting SPX implied volatilities does not automatically generate realistic VIX option prices.

This weakness appears clearly in the standard SABR VIX results. Although the model remains well calibrated on SPX, its performance on VIX options is poor, with very large discrepancies between model and market implied volatilities for several maturities. This is fully consistent with the global comparison reported later in the notebook: standard SABR achieves the best SPX fit but a very poor VIX fit. In order to address this limitation, the project therefore introduces a joint calibration procedure in which VIX information is incorporated directly into the objective function.
More precisely, the joint SABR calibration augments the SPX fitting criterion with a VIX penalty term. For each SPX maturity slice, the calibration may be matched with a nearby VIX maturity whenever the difference in maturities is sufficiently small. The joint loss then combines the SPX smile fitting error with a weighted VIX pricing error obtained from Monte Carlo. The purpose of this extension is not to preserve the same near-perfect SPX fit as in the standard version, but rather to investigate whether a better compromise between equity-option and volatility-option information can be achieved within the SABR framework.

The resulting comparison between standard and joint SABR on VIX options is particularly informative. The joint specification substantially improves the fit to VIX implied volatilities and corrects the most severe discrepancies observed under the standard calibration. At the same time, this improvement comes at a visible cost on the SPX side, where the standard SABR version remains superior. This trade-off is one of the central empirical conclusions of the project: within the SABR framework, introducing VIX information materially enhances the volatility-option fit, but this gain is obtained at the price of a deterioration in SPX accuracy.

VIX Calls : SABR Standard (rouge) vs SABR Joint (bleu) vs Marché

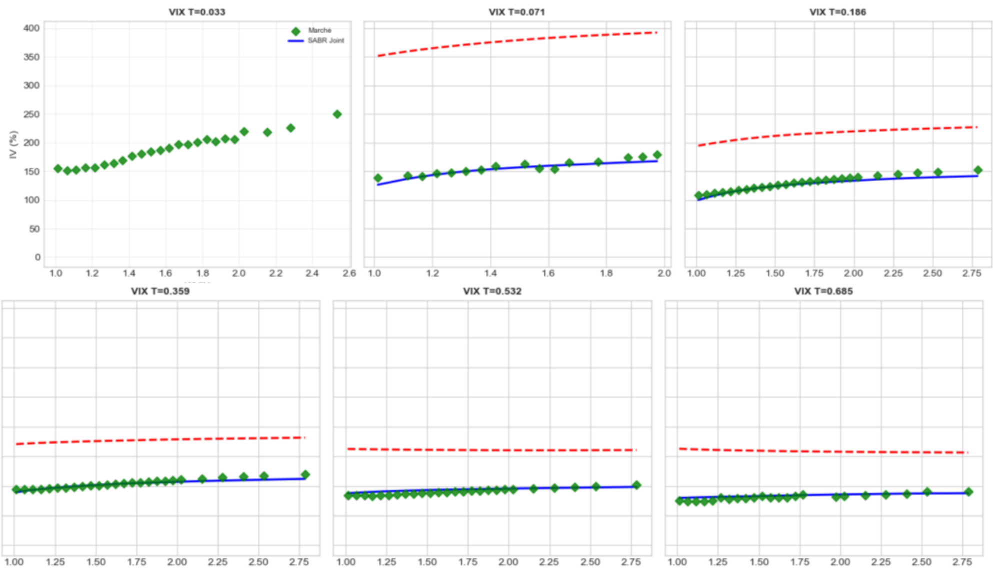


Figure 6. Comparison of market VIX implied volatilities with standard SABR and joint SABR implied volatilities across selected maturities. The figure shows the strong improvement generated by the joint calibration on the VIX side.

Overall, the SABR results highlight both the strengths and the limitations of the model. On the one hand, the standard calibration provides an excellent fit to SPX implied volatility smiles and therefore constitutes the strongest pure SPX performer in the project. On the other hand, the same specification struggles to reproduce VIX option information in a satisfactory way. The joint calibration substantially improves the VIX fit, but this comes with a clear degradation of SPX performance. The SABR model therefore emerges as a very effective local cross-sectional fitting tool, but a less convincing unified framework when one requires simultaneous consistency between equity-index options and volatility derivatives.

4. Heston Model

In the second part of the project, we consider the Heston stochastic volatility model (Heston, 1993) as an alternative framework for jointly analyzing SPX and VIX option data. Under the risk-neutral measure, the dynamics are written as

$$dS_t = (r-q)S_t\,dt + \sqrt{V_t}\,S_t\,dW_t^{(1)}, \qquad dV_t = \kappa(\theta - V_t)\,dt + \xi\sqrt{V_t}\,dW_t^{(2)},$$

with

$$d\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,dt.$$

Here, $V_t$denotes the instantaneous variance process, $\kappa$the mean-reversion speed, $\theta$the long-run variance level, $\xi$the volatility of variance, and $\rho$the instantaneous correlation between the asset and variance shocks. Unlike the slice-by-slice SABR implementation, Heston provides a genuine stochastic variance model with a unified parametric structure across maturities. This makes it particularly relevant when the objective is not only to fit SPX smiles, but also to preserve some consistency between equity-index options and volatility derivatives.

A key advantage of the Heston setup in the final notebook is that the model is used in two complementary ways. For SPX options, pricing is performed with the analytic Heston engine from QuantLib, which allows direct calibration on the cross-section of SPX implied volatilities. For VIX options, the project exploits the fact that under Heston the future VIX level can be related analytically to the future variance state. More precisely, conditionally on V_T, the squared VIX is given by

$$\mathrm{VIX}_T^2 = \theta + (V_T - \theta)\frac{1-e^{-\kappa\Delta}}{\kappa\Delta},$$

where Δcorresponds to the 30-calendar-day VIX window. In the retained implementation, only the terminal variance $V_T$has to be simulated, and the VIX level is then recovered through this exact analytical relation. This is an important difference relative to the SABR case, where VIX pricing is fully simulation-based and numerically noisier.

Before presenting the calibration results, it is useful to recall the nature of the Heston dynamics through simulated variance paths. In the notebook, the variance process is simulated with the quadratic-exponential scheme (Andersen, 2007), which is well suited to the CIR-type structure of the variance dynamics and preserves non-negativity more robustly than a naive Euler discretization. The following figure illustrates several sample paths of the model-implied volatility $\sigma(t) = \sqrt{V_t}$, together with the long-run volatility level $\sqrt{\theta}$. The figure highlights the mean-reverting behavior of the variance process, but also the possibility of large temporary volatility excursions, which is one of the reasons why the Heston framework remains attractive in practice.

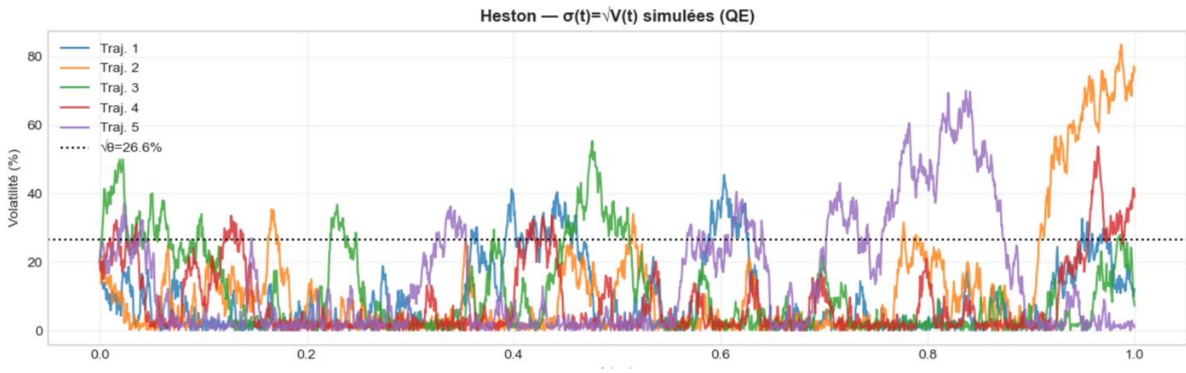


Figure 7. Simulated volatility paths under the Heston model using the quadratic-exponential scheme, with the long-run volatility level shown for reference.

The standard Heston calibration is then performed on the SPX option sample retained after preprocessing. In the final code, the calibration is carried out with QuantLib by minimizing vega-weighted residuals between model prices and market prices. The weighting scheme is important because it avoids over-emphasizing contracts with very low sensitivity and provides a more balanced fit across strikes and maturities. In addition, the SPX calibration is performed on the restricted SPX subsample defined in the notebook, which concentrates the fit on the most relevant region of the SPX smile and improves numerical stability.

The resulting SPX smiles show that Heston is able to capture the general term-structure and curvature of the equity implied volatility surface, but less accurately than standard SABR. This conclusion is consistent with the global RMSE comparison reported later in the notebook: Heston standard remains clearly above SABR standard in terms of SPX fitting error. In particular, for short maturities and for some wings of the smile, the Heston fit is visibly less precise than the slice-by-slice SABR adjustment. This difference is not surprising. SABR benefits from a highly flexible maturity-by-maturity calibration, whereas Heston imposes a single set of structural parameters across the entire SPX sample. The price paid for greater dynamic coherence is therefore a weaker local cross-sectional fit.

SPX Calls : Heston Standard vs Marché

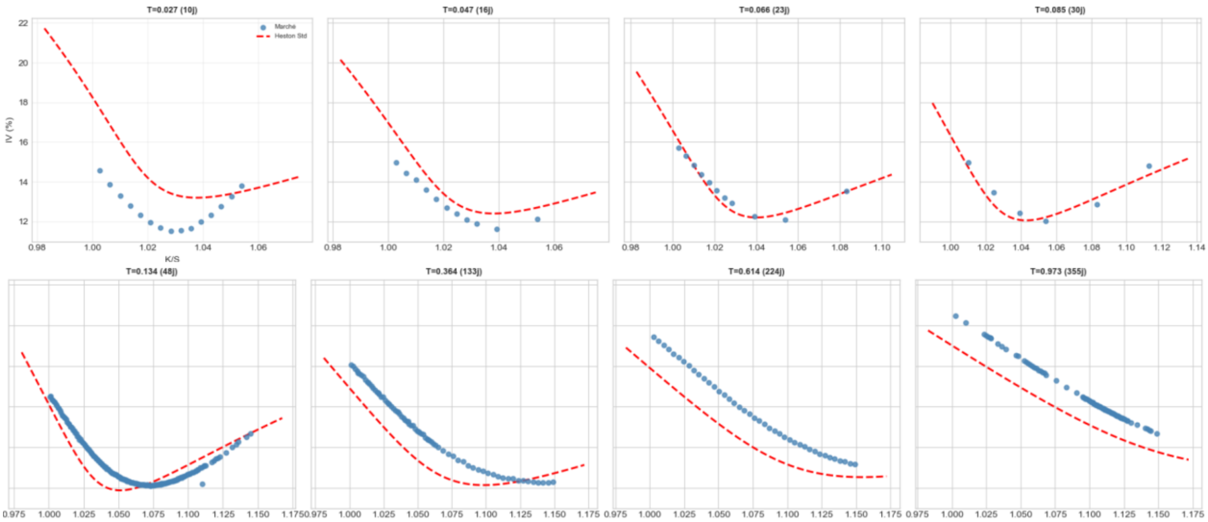
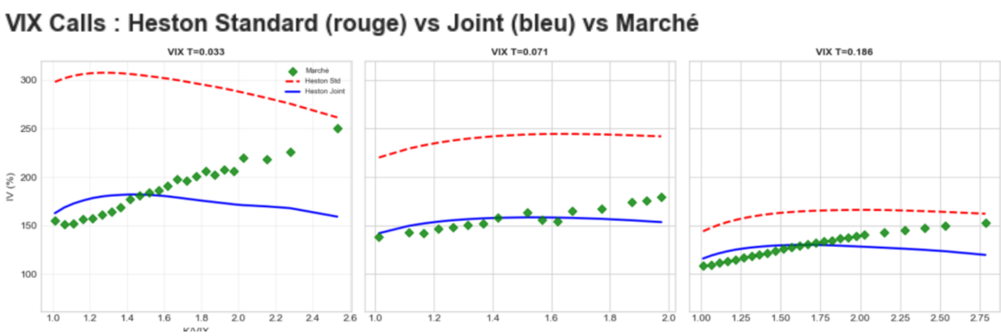


Figure 8. Market and Heston-implied volatility smiles for SPX out-of-the-money call options across selected maturities under the standard calibration.

The extension to VIX options is one of the most important contributions of the Heston part of the project. In the standard Heston version, VIX call prices are computed by simulating terminal variance values V_T under the quadratic-exponential scheme and then applying the exact analytical mapping from V_T to VIX_T. Relative to a full pathwise volatility simulation, this produces a numerically cleaner pricing framework and avoids introducing an additional discretization bias over the VIX window. Nevertheless, a standard Heston calibration based only on SPX options does not automatically produce a satisfactory VIX fit. In the final results, Heston standard performs better than standard SABR on VIX options, but it still remains far from the best achievable fit.

To improve this aspect, the notebook introduces a joint Heston calibration in which VIX information is incorporated directly into the objective function. The SPX component of the loss remains based on vega-weighted QuantLib pricing residuals, while the VIX component compares model and market VIX option prices using the exact analytical relation between VIX and future variance. This is a major numerical advantage of the Heston framework in the present study. Because the VIX residuals do not rely on a noisy full-path Monte Carlo construction of future integrated volatility, the optimization landscape remains substantially smoother than in the SABR joint case.

The comparison between standard and joint Heston on VIX options shows a clear improvement after joint calibration. The blue curves are generally closer to market implied volatilities than the red curves, which indicates that adding VIX information materially improves the model's ability to account for volatility-derivative prices. At the same time, this improvement does not lead to the same kind of extreme SPX–VIX trade-off observed under SABR. According to the final notebook results, Heston joint slightly improves the VIX fit relative to Heston standard while leaving the SPX fit in the same order of magnitude. This makes joint Heston a more balanced specification overall, even if it still does not match the pure SPX accuracy of standard SABR.



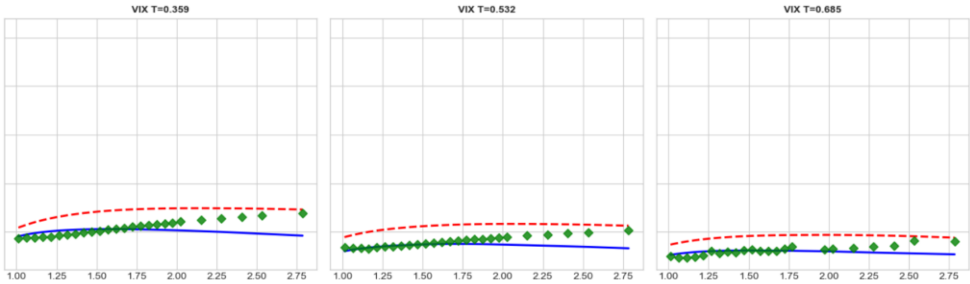


Figure 10. Market, standard Heston, and joint Heston implied volatilities for VIX out-of-the-money call options across selected maturities.

Unlike SABR, Heston produces a single structural parameter vector for each calibration. The calibrated standard and joint parameter values are reported in Table 2.

Calibration	**(\v_0)**	**(\kappa)**	**(\theta)**	**(\xi)**	**(\rho)**
Heston Standard	0.040286	3.890705	0.070604	2.123738	-0.786021
Heston Joint	0.029789	2.868117	0.060745	1.246779	-0.749871

Table 2. Calibrated Heston parameters

Overall, the Heston results complement the SABR analysis in an economically meaningful way. Standard Heston is less accurate than standard SABR on SPX options, but it provides a smoother and more structurally coherent description of the full volatility surface. More importantly, once VIX information is introduced, the Heston framework appears more naturally suited to joint SPX–VIX modeling. The availability of an exact analytical relation between the future variance state and the VIX level is central here: it makes VIX pricing more tractable and supports a better numerical behavior of the joint calibration. In the final comparison of the project, Heston therefore emerges not as the best pure SPX fitter, but as the most convincing compromise when one seeks a model that can incorporate both equity-option and volatility-option information within a single stochastic volatility framework.

5. Hölder Regularity

Beyond calibration quality, the project also examines the path regularity of the simulated processes generated by the two models. The purpose of this final numerical exercise is not to recalibrate the models, but rather to verify whether their simulated trajectories are consistent with the regularity expected from standard diffusion processes. To do so, we estimate empirically a Hölder exponent Hthrough the scaling relation

$$\mathbb{E}\left[|X_{t+h} - X_t|\right] \sim h^H \qquad \text{as } h \to 0.$$

For a Brownian diffusion, one expects $H \approx 0.5$. By contrast, rough-volatility models are typically associated with much smaller values, often in the range 0.05to 0.15. The objective here is therefore to assess whether the simulated dynamics of SABR and Heston behave as standard Brownian diffusions or display a significantly rougher path structure.

The empirical procedure implemented in the notebook is deliberately simple and fully consistent across models. For a sequence of time discretizations, simulated paths are generated over a one-year horizon and the mean absolute increment $\mathbb{E}[\mid \Delta X \mid]$is computed. A log-log regression of $\log(\mathbb{E}[\mid \Delta X \mid])$on $\log(\Delta t)$then provides an estimate of the slope, which is interpreted as the empirical Hölder exponent. In the SABR case, the exercise is performed on the forward process $F_t$and on the volatility process σ_t, using representative parameter sets reported in Table 1. In the Heston case, the same methodology is applied to the variance process V_t, using the calibrated standard and joint parameter values. In all cases, the path simulations are based on the final implementation retained in the notebook.

For SABR, the log-log plots confirm that the simulated processes are very close to the theoretical Brownian benchmark. The estimated exponent for the standard SABR forward process is approximately $H \approx 0.497$, while the corresponding value for the joint SABR forward process is $H \approx 0.495$. For the volatility process, the estimated exponent is about $H \approx 0.480$under standard SABR and $H \approx 0.492$under joint SABR. These values are all close to 0.5, and the fitted lines remain visually very near the theoretical reference slope. The conclusion is therefore clear: in the present implementation, the SABR dynamics behave as standard diffusion-type processes from the point of view of path regularity.

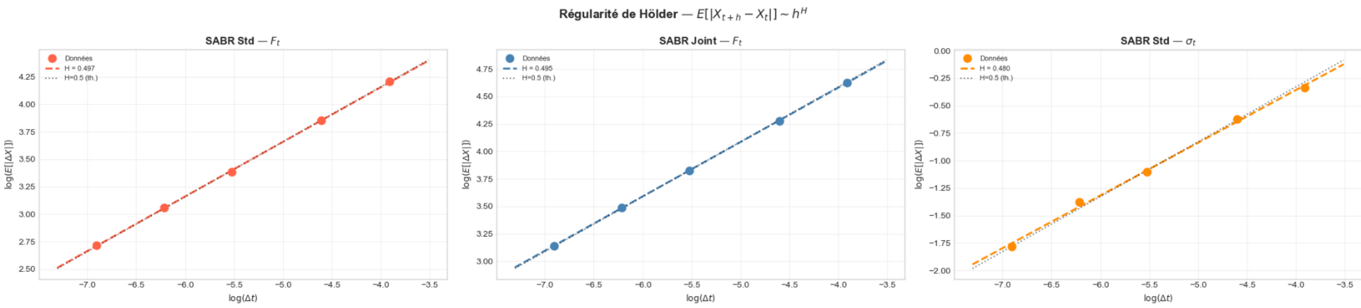


Figure 11. Log-log estimation of the empirical Hölder exponent under the SABR model for the forward process and the volatility process.

The Heston results lead to the same conclusion. The estimated exponent for the standard Heston variance process is approximately $H \approx 0.491$, while the joint Heston variance process yields $H \approx 0.490$. Once again, these estimates are extremely close to the theoretical value 0.5, and the fitted lines are almost indistinguishable from the Brownian benchmark in the log-log representation. This confirms that the Heston variance process, as simulated in the notebook, also exhibits the regularity expected from a classical stochastic diffusion.

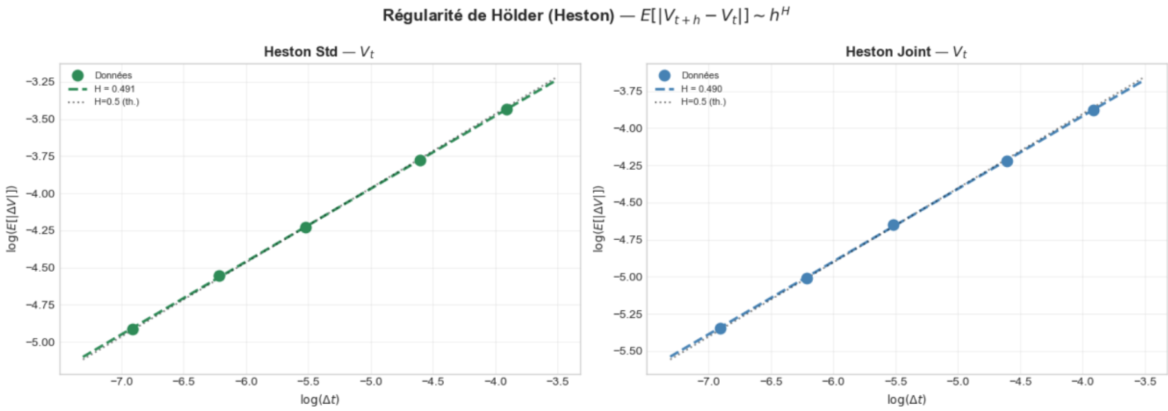


Figure 12. Log-log estimation of the empirical Hölder exponent under the Heston model for the variance process.

These results are important for the interpretation of the whole project. They show that, despite their differences in calibration structure and in their ability to fit SPX and VIX options, both SABR and Heston remain standard Brownian stochastic-volatility models in terms of path regularity. In other words, neither of the two frameworks generates the kind of rough trajectories that have been emphasized in the rough-volatility literature. This point is fully consistent with theory: both models are driven by standard Brownian motions and are therefore not designed to reproduce very low Hölder exponents.

From an economic point of view, this regularity analysis also helps clarify the scope of the modeling exercise. The empirical comparison carried out in the previous sections concerns the ability of the models to fit observed option prices and implied volatilities, especially across SPX and VIX markets.

The Hölder analysis shows that these calibration differences do not come from fundamentally different path regularities, but rather from the structural characteristics of the models themselves. Standard SABR, joint SABR, standard Heston, and joint Heston all remain diffusion-based specifications with empirical exponents close to 0.5. Therefore, if one wished to capture genuinely rough volatility behavior, one would need to move beyond the present frameworks toward models specifically designed for that purpose.

Overall, the regularity exercise confirms the internal consistency of the final notebook. The estimated Hölder exponents remain close to the theoretical diffusion value in every case, which supports the interpretation of both SABR and Heston as classical stochastic-volatility models rather than rough-volatility models. This final result complements the calibration analysis by showing that the trade-offs observed earlier between SPX fit, VIX fit, and joint calibration performance occur within a common diffusion-based regularity class.

Model	**Process**	**H (estimated)**	**Theoretical**
SABR Standard	F_t (forward)	0.4968	0.500
SABR Standard	σ_t (volatility)	0.4800	0.500
SABR Joint	F_t (forward)	0.4946	0.500
SABR Joint	σ_t (volatility)	0.4917	0.500
Heston Standard	V_t (variance)	0.4905	0.500
Heston Joint	V_t (variance)	0.4896	0.500

Table 3. Empirical Hölder exponents estimated from 5,000 simulated paths over a one-year horizon (log-log regression).

6. Global Comparison and Discussion

The final stage of the project consists in comparing the four retained specifications on a common empirical basis. The purpose is not only to rank the models according to a single error metric, but also to understand the trade-off between SPX fitting accuracy and VIX consistency that emerges from the calibration results. In this respect, the comparison confirms that no single specification dominates in every dimension. Instead, the results reveal a clear opposition between local SPX fit and cross-market robustness.

Model	**RMSE SPX**	**MAE SPX**	**RMSE VIX**	**Calib. (s)**	**Params**
SABR Standard (β*=0.7)	0.429	0.114	117.051	2.3	$\beta + 3\times N_{mat}$
SABR Joint	0.890	—	14.690	449	$\beta + 3\times N_{mat}$
Heston Standard	1.067	0.930	57.746	19.4	5
Heston Joint (analytic VIX)	1.029	0.852	15.577	32.7	5

Table 4. Global RMSE and MAE summary across all four calibrations (RMSE in implied volatility points).

The global RMSE comparison is summarized in the following table. The standard SABR calibration achieves the best SPX fit, with an SPX RMSE of 0.429volatility points. This is the lowest value among all specifications and confirms the conclusions already drawn from the SPX smile figures: as a slice-by-slice fitting device, standard SABR is extremely effective on the equity-index option surface. However, this strong SPX performance comes together with a very poor VIX fit. The corresponding VIX RMSE reaches 117.051volatility points, by far the largest error in the study. In other words, the standard SABR specification is the best pure SPX model in the project, but it fails to provide a satisfactory joint description of SPX and VIX option information.

By construction, the joint SABR calibration attempts to correct this weakness by incorporating VIX information directly into the objective function. The effect is very visible in the RMSE values: the VIX RMSE falls sharply from 117.051to 14.690volatility points. This is the best VIX result obtained in the whole project. The cost of this gain is equally clear on the SPX side, where the RMSE increases from 0.429to 0.890. Although the joint version remains acceptable in absolute terms, it is much less accurate than standard SABR on the equity surface. The empirical message is therefore straightforward: within the SABR framework, it is possible to fit VIX options much better, but only by giving up a significant part of the excellent SPX fit delivered by the standard calibration.
The Heston results exhibit a different pattern. Standard Heston produces an SPX RMSE of 1.067, which is clearly above both standard and joint SABR. This confirms that, in the present implementation, Heston is less flexible than SABR for matching the SPX implied volatility cross-section. At the same time, standard Heston performs much better than standard SABR on VIX options, with a VIX RMSE of 57.746. This is still far from the best VIX result, but it is already substantially more satisfactory than the standard SABR outcome. Hence, even before introducing VIX information explicitly into the calibration, Heston appears more naturally compatible with volatility-derivative pricing.

The joint Heston calibration further improves this balance. In the final results, the SPX RMSE decreases slightly from 1.067to 1.029, while the VIX RMSE falls from 57.746to 15.577. The gain on the VIX side is therefore large, and the SPX fit remains in the same general range as under standard Heston. This behavior contrasts with the sharper SPX–VIX trade-off observed under SABR. Although joint Heston does not reach the SPX precision of standard SABR, and its VIX fit remains slightly worse than joint SABR, it delivers the most balanced overall outcome among the Heston specifications and one of the most convincing compromises in the project as a whole.

Modèle	RMSE SPX	RMSE VIX
SABR Standard	**0.429**	117.051
SABR Joint	0.890	**14.690**
Heston Standard	1.067	57.746
Heston Joint	1.029	15.577

Figure 13. Global comparison of SPX and VIX RMSE across the four calibrated model specifications.

Taken together, these results suggest a natural interpretation of the four model variants. Standard SABR should be viewed as the strongest specification when the objective is to reproduce the SPX implied volatility surface as accurately as possible. Joint SABR should be viewed as the best performer on VIX options, but at the price of a clear degradation in SPX fit. Standard Heston occupies an intermediate position, with weaker SPX accuracy than SABR but a noticeably more reasonable VIX performance. Joint Heston, finally, appears to offer the best global compromise: it improves VIX pricing substantially while preserving a relatively stable SPX fit. In the context of this project, the model choice therefore depends on the modeling priority. If the emphasis is placed on SPX surface

fitting alone, standard SABR is the natural candidate. If one seeks a more balanced joint SPX–VIX framework, joint Heston is the most convincing option.
This interpretation is reinforced by the term-structure-of-skew comparison on SPX options. The skew figure shows that the market skew declines with maturity over the range considered in the notebook. Standard SABR tracks this pattern relatively well, especially over the central part of the maturity interval, which is consistent with its strong SPX RMSE performance. Heston standard, by contrast, tends to generate a steeper skew level than observed in the market, particularly at shorter maturities. Joint Heston produces a flatter curve, generally below the market skew, but still captures the decreasing maturity pattern. The figure is therefore useful because it reveals differences that are not fully visible from RMSE values alone: even when two models have similar average errors, they may differ substantially in the way they reproduce the economic shape of the smile across maturities.

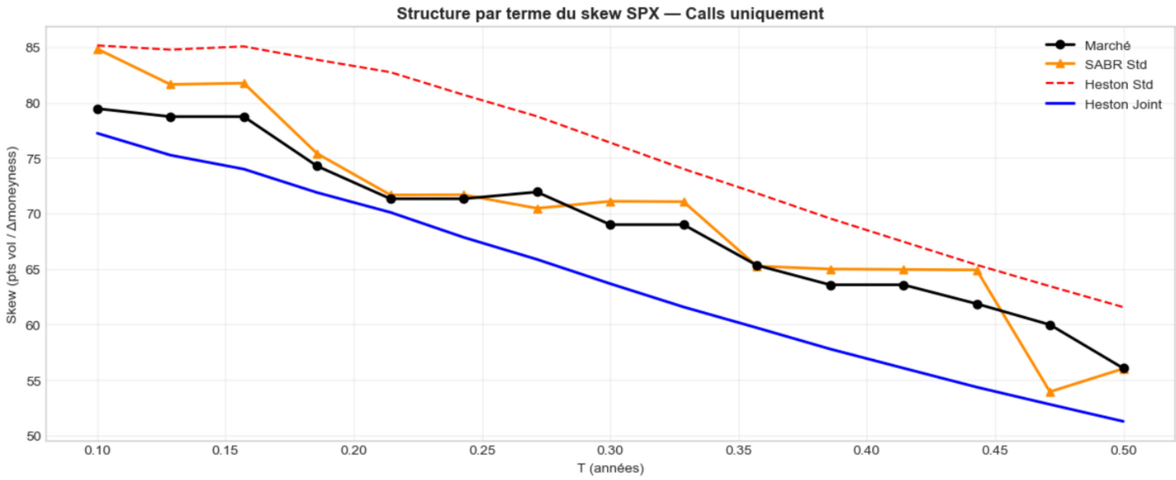


Figure 14. Term structure of SPX skew in the market and under the different calibrated model specifications.

From a broader perspective, the comparison highlights the central tension of the project. A model that is highly effective as a cross-sectional fitting tool on SPX options is not necessarily the most suitable one for incorporating VIX information. Conversely, a model with a more coherent stochastic-volatility structure may provide a weaker local SPX fit but a more satisfactory joint treatment of equity and volatility derivatives. This is exactly what the final results show. SABR excels as a local smile-fitting model, whereas Heston appears more robust when the analysis is extended to VIX options and joint calibration.

A further practical consideration concerns computational efficiency. Table 4 reveals a striking asymmetry in calibration time: the SABR joint calibration requires approximately 449 seconds, compared with only 32.7 seconds for the Heston joint calibration. This difference, amounting to a factor of roughly 14, stems from the Monte Carlo pricing of VIX options under SABR, which involves a full pathwise simulation of the volatility process for each evaluation of the joint objective. By contrast, Heston exploits the exact analytical relation between the terminal variance V_T and the VIX level, so the VIX residuals are computed quasi-analytically and the optimization landscape is substantially smoother. This computational advantage of the Heston joint specification is an important practical argument in its favor, beyond its already better-balanced RMSE profile.

Overall, the empirical findings of the project can be summarized as follows. Standard SABR provides the best SPX fit by a clear margin. Joint SABR delivers the best VIX fit, but only by sacrificing a significant part of the SPX accuracy. Standard Heston is less competitive on SPX alone, yet more reasonable on VIX than standard SABR. Joint Heston achieves the most balanced global performance, combining a substantial improvement in VIX fit with a relatively stable SPX fit. In that sense, the final comparison does not identify a universally best model, but rather a hierarchy of specifications adapted to different modeling objectives.

7. Conclusion

This project compared the SABR and Heston stochastic volatility frameworks on a fixed market snapshot of SPX and VIX out-of-the-money call options. For each model, we considered both a standard calibration, mainly based on SPX information, and a joint calibration incorporating VIX option data. We also examined the empirical Hölder regularity of the simulated processes in order to relate the numerical results to the theoretical diffusion structure of the models.

The results show that model performance depends strongly on the calibration objective. When the priority is to fit the SPX implied volatility surface as accurately as possible, standard SABR is the best performer. Its slice-by-slice calibration produces the lowest SPX RMSE, but this strong local fit does not extend to VIX options, where the model performs poorly. Joint SABR corrects this weakness substantially: it delivers the best VIX fit in the project, but at the cost of a clear deterioration in SPX fit.

Heston leads to a different conclusion. Standard Heston is less accurate than standard SABR on SPX options, but it behaves more reasonably on VIX options even before joint calibration. Once VIX information is added, the Heston fit improves significantly on the VIX side while keeping the SPX fit in the same general range. In this sense, joint Heston provides the most balanced overall specification of the study, even if it does not match standard SABR on SPX alone or joint SABR on VIX alone. The regularity analysis complements these findings. For all retained specifications, the estimated Hölder exponents remain close to 0.5, which confirms that both SABR and Heston behave as standard Brownian diffusion models from the point of view of path regularity. The observed differences in calibration performance therefore come from the structural properties of the models and from the way SPX and VIX information enter the calibration procedures, rather than from different regularity classes.

The study nevertheless has some limitations. VIX implied volatilities are constructed using the spot VIX as a proxy for the relevant VIX futures underlying, which is a simplifying assumption. In addition, the analysis is based on a single market snapshot and should therefore be interpreted as a cross-sectional comparison rather than a universal model ranking. Overall, the project highlights a clear trade-off in volatility modeling: standard SABR is the best choice for fitting the SPX surface alone, joint SABR is the strongest on VIX options, and joint Heston emerges as the most balanced framework when both SPX and VIX information are taken into account.

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